

llmXive Automated Review of PlanBench-XL: Evaluating Long-Horizon Planning of LLM Tool-Use Agents in Large-Scale Tool Ecosystems

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper makes several strong factual claims regarding model performance drops and statistical correlations that require precise alignment with the reported data tables and specific experimental settings.

First, there is a conflation of experimental conditions in the narrative description of performance drops. The Abstract and Section 4.2 state that GPT-5.4 accuracy drops to “around 30%” when “only one feasible solution path remains,” but Section 5.1 explicitly distinguishes between a “Mixed” setting (where one path remains, yielding ~30%) and a “Longest path” setting (yielding ~11.36%). The text in Section 4.2 appears to merge these, stating GPT-5.4 falls to “around 30% when only one feasible path remains and to slightly above 10% when only the longest recovery path is preserved.” However, the Abstract simplifies this to a single “severe blocking” figure of 11.36%, which only applies to the specific “Longest path” constraint. This lack of distinction in the high-level summary obscures the nuance that the drop to 11.36% is specific to the longest-path constraint, not just “one path remaining.”

Second, the claim of a strong Pearson correlation ($r = 0.902$) between Mean Explored Datatypes (Mean EDT) and Accuracy in Section 4.2 warrants scrutiny. While the top-performing models (Gemini-3.1-Pro, GPT-5.4, DeepSeek-V4-Flash) show a positive trend, the inclusion of models with 0% accuracy (Qwen3-8B, Llama-3.1-8B) which still exhibit non-zero Mean EDT (7.64 and 9.89 respectively) suggests the relationship is not strictly linear across the entire dataset. The high correlation coefficient may be driven disproportionately by the variance among the top three models. The claim that exploration tendency “strongly relates” to success is valid, but the specific $r = 0.902$ figure might overstate the predictive power of exploration for the lower-performing models that explore but fail to execute correctly.

Finally, the specific accuracy figure of 11.36% cited in the Abstract for “severe blocking” is only accurate for the “Longest path” scenario described in Figure 5 and Section 5.1. In the “Mixed” blocking scenario (also with one path remaining), the accuracy is significantly higher (~30%). The Abstract should explicitly qualify that the 11.36% figure corresponds to the “longest path” constraint to avoid misrepresenting the general difficulty of the “one path” condition.

Action items.

- [writing] The abstract and Section 4.2 conflate two blocking scenarios. GPT-5.4 drops to ~30% in the ‘Mixed’ setting (one path) but to 11.36% in the ‘Longest path’ setting. The text ambiguously attributes the 11.36% drop to ‘severe blocking’ generally, obscuring that this specific low score applies only to the longest-path constraint.

- [science] Section 4.2 claims a Pearson $r=0.902$ correlation between Mean EDT and Accuracy. However, models with 0% accuracy (Qwen3-8B, Llama-3.1-8B) still show non-zero exploration (7.64, 9.89 EDT). The high correlation may be driven solely by the top 3 models, potentially overstating the linear relationship across the full performance range.
- [writing] The abstract states GPT-5.4 drops to 11.36% under ‘severe blocking’. This figure specifically applies to the ‘Longest path’ setting (Section 5.1), not the general ‘one path’ scenario which yields ~30%. The abstract must specify the ‘Longest path’ constraint to avoid misrepresenting the general difficulty of the blocking condition.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure clearly displays the annotation interface described in the caption, showing a tool document, input schema, and a 1–5 Likert scale for rating tool reasonableness. All elements are legible and consistent with the stated purpose.

Figure 2

The figure clearly displays the annotation interface for evaluating datatype quality, showing the ‘payment_status’ definition and the 1–5 Likert scale rating options. The visual content aligns perfectly with the caption’s description of the task and scale.

Figure 3

The figure provides a clear and comprehensive visual overview of the data construction and interaction protocol. However, the caption contains a grammatical error (‘Overview of .’), and the diagram itself has a spelling typo (‘Solvabe’).

Figure 4

The figure presents accuracy trends under varying enforced budgets, but inconsistent y-axis scales across subplots distort the visual comparison of performance gains, and the y-axis label is not clearly associated with all panels.

Figure 5

The figure is visually clear with a readable legend and error bars, but the caption contains a grammatical error ('on under'), and the specific model names on the x-axis contradict the abbreviations used in the rest of the paper.

Figure 6

The figure presents a failure category distribution, but the stacked bars for several models do not sum to 100%, contradicting the 'Distribution' claim in the caption. Additionally, the visual segments do not consistently match the four categories defined in the legend.

Figure 7

Figure 7 visually maps a trajectory with labeled steps and metrics, but the caption's claim of '10 explored datatypes' conflicts with the 10 tool calls shown, and the origin of the 'noisy tool' label lacks contextual clarity for metric derivation.

Figure 8

The figure contains a significant structural contradiction where the left panel's x-axis labels list model names instead of the block types described in the caption, confusing the variable being compared. Additionally, the rotated x-axis labels in the left panel reduce readability.

Figure 9

The figure presents a stacked bar chart of block alternative distributions, but the percentage labels for DeepSeek and Gemini do not sum to 100% despite a visible 'Misleading' segment, and the absence of this segment in GPT/Llama bars without explanation creates ambiguity.

Figure 10

Figure 10 is critically flawed due to a complete mismatch between the legend colors and the bar colors, and a contradiction between the caption's description of 'row length' and the vertical bar chart format. The x-axis also fails to distinguish between the 'block types' mentioned in the caption.

Figure 11

The figure effectively demonstrates low variance across seeds for both models as claimed in the caption. However, the lack of numerical data labels or error bars makes it difficult to independently verify the specific '3%' variation threshold mentioned in the text.

Figure 12

The figure effectively illustrates the decline in accuracy with path length, but panel (b) appears to be missing the data bars for the ' $L^* \geq 8$ ' category defined in the legend, and the x-axis label rotation is suboptimal.

Action items.

- [writing] Figure 3: The caption begins with 'Overview of ' which contains a grammatical error and missing noun phrase (likely 'the framework' or 'PlanBench-XL').
- [writing] Figure 3: The diagram contains a typo in the bottom arrow label, spelling 'Solvabe' instead of 'Solvable'.
- [science] Figure 4: The y-axis scales are inconsistent across subplots (ranging from ~12.5% to ~60%) without explicit indication, making visual comparison of 'limited accuracy gains' misleading.
- [writing] Figure 4: The y-axis label 'Accuracy (%)' is placed only on the far left; it should be repeated or clarified for the other three subplots to avoid ambiguity.
- [writing] Figure 5: The caption reads 'Accuracy (%) on under fine-grained blocking,' which contains a grammatical error and missing noun (likely 'on PlanBench-XL' or similar).
- [science] Figure 5: The x-axis labels (e.g., 'GPT-5.4', 'Gemini-3.5-Flash') appear to be hallucinated or placeholder model names that do not match the standard abbreviations (GPT, Gemini, DeepSeek, Llama) defined in the captions for Figures 6, 9, and 10, creating a potential inconsistency in model identification.
- [science] Figure 6: The stacked bars for GPT, DeepSeek, and Gemini do not sum to 100% (e.g., GPT sums to ~86.2%), yet the caption describes a 'Distribution' under a default setting, implying a complete breakdown of trajectory outcomes. The missing percentage is not accounted for in the legend or labels.
- [writing] Figure 6: The legend defines four categories ('No Traction', 'Irrecoverable Drift', 'Weak Recovery', 'Format Error'), but the bars for GPT and Gemini visually display only three segments, making it unclear which category is absent or if the data is incomplete.
- [writing] Figure 7: The caption states the trajectory yields '10 explored datatypes' (Turns = 10), but the diagram explicitly labels only 10 tool calls (steps 2-10). The distinction between 'turns' and 'explored datatypes' is not visually represented or defined in the figure or caption, making the metric calculation opaque.
- [science] Figure 7: The 'noisy off-path' arrow (6) originates from 'order id' and points to 'return request (untrusted)', yet the label says 'tool2 (noisy tool)'. However, step (3) in the top retrieval box lists 'tool2, tool3' as retrieved together, but the diagram doesn't clarify why tool2 is noisy while tool3 is used on-path — this ambiguity undermines the metric's reproducibility.

- [science] Figure 8: The x-axis labels in the left panel ('GPT-5.4', 'Gemini-3.5-Flash', 'DeepSeek-V4-Flash', 'Llama-3.3-70B-Instruct') contradict the caption, which states the left panel shows 'performance across different block types' (Mixed, Implicit, Explicit, Misleading). The current labels suggest the x-axis represents models, while the legend represents block types, reversing the intended comparison structure described in the text.
- [writing] Figure 8: The x-axis labels in the left panel are rotated at a steep angle, making them difficult to read and visually cluttered; horizontal alignment or a different layout would improve legibility.
- [science] Figure 9: The bars for GPT and Llama do not sum to 100% (GPT: $29.3\% + 70.7\% = 100\%$; Llama: $15.3\% + 84.7\% = 100\%$), but the legend includes a 'Misleading' category (dark teal) which is not visible in these bars, implying the data is incomplete or the legend is misleading for these specific models.
- [science] Figure 9: The bars for DeepSeek and Gemini show a small dark teal segment on the far right corresponding to 'Misleading', but the percentage labels ($48.6\% + 49.0\% = 97.6\%$ and $44.8\% + 52.2\% = 97.0\%$) do not account for this visible segment, making the data representation mathematically inconsistent.
- [fatal] Figure 10: The legend defines three categories (Unused, Search Reused, Value Reused) with dark, medium, and light gray colors, but the chart bars are colored green, yellow, and blue. The visual data does not match the legend or the caption's description of 'colored segments'.
- [fatal] Figure 10: The caption states 'Each column corresponds to one model and one block type', yet the x-axis labels show only model names (GPT, DeepSeek, Gemini, Llama) with no indication of the block type, making the data grouping ambiguous.
- [science] Figure 10: The y-axis is labeled 'Percentage (%)' and the bars reach 100, but the caption claims 'The row length shows the ratio', implying a horizontal bar chart. The mismatch between the vertical bar orientation and the 'row length' description creates confusion about what the 100% total represents.
- [science] Figure 11: The caption claims variations do not exceed 3%, but the visual difference between the highest bar (Seed 2000) and lowest bar (Seed 3000) for DeepSeek-V4-Flash appears to be approximately 1-2%, which is consistent. However, for Llama-3.3-70B-Instruct, the difference between Seed 1000/2000 and Seed 3000 is visually negligible (<1%). The claim is supported, but the chart lacks error bars or numerical labels to verify the specific '3%' threshold mentioned in the text.
- [writing] Figure 11: The x-axis labels ('DeepSeek-V4-Flash', 'Llama-3.3-70B-Instruct') are italicized, which is non-standard for model names in scientific figures and may reduce readability.
- [writing] Figure 12: The x-axis labels in panel (b) 'Block Setting' are missing the bars for the 'L* >= 8' group (light blue), yet the legend defines this category. The caption

mentions aggregating longer tasks into this group, but the data is visually absent in the right panel.

- [writing] Figure 12: The x-axis labels are rotated at a steep angle, which is unnecessary given the short text length and reduces readability compared to a horizontal or slight tilt.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology and undefined acronyms that hinder accessibility for non-specialist readers.

First, the core concept of “bi-directional anticipation” (Introduction, Section 3) is introduced without a plain-English definition. While the text later explains it as “forward” and “backward” search, the initial label is jargon-heavy. The authors should explicitly state: “We call this process bi-directional anticipation, where agents search forward from evidence and backward from goals.”

Second, the paper introduces a dense cluster of acronyms in Section 4.1 (EGT, EDT, S/C, ITCR, UIRR) without defining them in the main text before they appear in Table 2. For instance, “Executed Ground-Truth Datatype Precision” is a mouthful; the acronym EGT is used immediately in the table header without a prior sentence defining it. Similarly, “Untrusted Input Rejection Rate” (UIRR) appears in the abstract and results without a clear, simple definition of what constitutes an “untrusted input” for a general audience.

Third, the term “datatypes” is consistently used in CamelCase (e.g., `person_name`, `purchase_status`) throughout the text and figures. This is a programming convention that feels out of place in a natural language paper. Replacing this with “data types” or “information categories” would improve readability.

Finally, the concept of “blocking” is used as a technical noun (“retrieval-time blocking”) without a simple analogy. A brief explanation like “simulating missing or broken tools” would make the mechanism immediately clear to readers outside the specific sub-field of tool-use agents. The current phrasing assumes the reader already understands the specific failure modes being simulated.

Action items.

- [writing] The manuscript relies heavily on specialized terminology and undefined acronyms that hinder accessibility for non-specialist readers. First, the core concept of “bi-directional anticipation” (Introduction, Section 3) is introduced without a plain-English definition. While the text later explains it as “forward” and “backward” search, the initial label is jargon-heavy. The authors should explicitly state: “We call this process

bi-directional anticipation, where agents search forward from evidence

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically sound framework for evaluating LLM agents in large-scale tool ecosystems, but there are minor inconsistencies in how specific quantitative claims are linked to experimental conditions.

First, the abstract states that GPT-5.4 accuracy drops to 11.36% under “severe blocking.” However, Section 5 (Analysis) and Figure 5 describe the “1 Path” (longest path) setting as the most severe condition, where accuracy drops to “slightly above 10%.” While 11.36% is consistent with “slightly above 10%,” the text does not explicitly define “severe blocking” as the “1 Path” setting or the specific block ratio that yields 11.36%. This creates a slight disconnect between the abstract’s precise claim and the body’s qualitative description, making it difficult for a reader to immediately verify the exact condition associated with that number.

Second, the claim in Section 4.2 that Mean Explored Datatypes (Mean EDT) is strongly correlated with accuracy (Pearson $r=0.902$) is logically sound in direction but potentially overstated in precision given the data. The table shows models with similar Mean EDT (e.g., Llama-3.3-70B at 19.20 and GPT-5.4 at 20.65) having drastically different accuracies (18.96% vs 51.90%). While a high correlation can exist despite such outliers, the text does not address this discrepancy or discuss whether the correlation holds when excluding specific models. This weakens the causal implication that “exploration tendency” alone explains performance, as the text admits other factors (like EGT Precision) are also critical.

Finally, the definition of “blocking” in the abstract (“severe blocking”) is not rigorously defined in the main text as a specific parameter (e.g., block ratio = 0.8 or “1 Path” constraint). The paper uses “severe blocking” as a narrative label for the “1 Path” or high-ratio settings, but the lack of a precise mapping in the abstract to the specific experimental setup in Section 5 creates a minor logical gap in reproducibility of the claim.

These issues are primarily presentational and do not invalidate the core findings, but clarifying the exact mapping between “severe blocking” and the specific experimental settings (block ratio/path constraint) would strengthen the logical consistency of the claims.

Action items.

- [writing] The abstract claims GPT-5.4 accuracy drops to 11.36% under “severe blocking,” but Section 5 describes the “1 Path” setting as yielding “slightly above 10%.” The text fails to explicitly link the specific value 11.36% to a defined blocking condition (e.g., specific block ratio or path constraint), creating ambiguity between the abstract’s precise claim

and the results section’s qualitative description.

- [science] Section 4.2 claims a Pearson correlation of $r=0.902$ between Mean EDT and accuracy. However, Table 3 shows models with similar Mean EDT (e.g., Llama-3.3-70B at 19.20 vs GPT-5.4 at 20.65) having vastly different accuracies (18.96% vs 51.90%). The text does not discuss how such outliers affect the correlation or whether the relationship is robust, weakening the causal implication that exploration alone drives performance.
- [writing] The term “severe blocking” is used in the abstract to justify the 11.36% figure, but the main text defines blocking conditions by “block ratio” or “path constraint” (e.g., “1 Path”). The paper lacks a precise definition mapping “severe blocking” to a specific experimental parameter, making the causal claim difficult to verify without assuming the “1 Path” setting.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding the capabilities of the benchmark and the nature of agent failures that slightly overreach the presented evidence.

First, the abstract and conclusion characterize the performance drop of GPT-5.4 to ~11% as a result of “severe blocking.” However, the data in Section 5 (Figure 5 and Figure 6) clarifies that this specific low accuracy is observed primarily when the blocking strategy preserves *only the longest* valid solution path, or under specific “Mixed” block types. Other blocking configurations (e.g., preserving the shortest path or using explicit errors) result in significantly higher accuracy (e.g., ~30%). By generalizing the “longest path” scenario to “severe blocking” without qualification, the paper overstates the universality of the 11% failure rate. The claim should be refined to specify that the most dramatic degradation occurs when recovery requires navigating the longest alternative paths.

Second, the paper introduces “bi-directional anticipation” (forward and backward retrieval) as a defining feature of the benchmark’s design. While the benchmark *supports* this, the results in Section 4 suggest that current models largely fail to engage in backward anticipation, relying instead on forward exploration. The paper claims the benchmark “tests” this capability, but the evidence primarily shows that models *lack* this capability. The overreach lies in implying the benchmark successfully elicits bi-directional planning, whereas the data suggests it primarily exposes the absence of such planning in current agents. The framing should be adjusted to reflect that the benchmark *reveals* the inability of agents to effectively utilize bi-directional strategies, rather than successfully testing a capability they do not possess.

Finally, the claim that “failures are most common when errors lack explicit signals” is

supported by accuracy metrics (implicit failures yield lower accuracy) but not by a direct analysis of failure *frequency* distribution. The paper does not explicitly state whether implicit failures occur more often than explicit ones, or if they simply cause more catastrophic outcomes when they do occur. Without a breakdown of the prevalence of each error type in the failure set, the claim that they are “most common” is an extrapolation from the severity of the outcome rather than the frequency of the event.

Action items.

- [writing] The abstract claims GPT-5.4 drops to 11.36% under ‘severe blocking,’ but this specific low accuracy only occurs when the longest recovery path is preserved (Section 5). Other block types yield ~30%. Clarify that the 11% figure is specific to the longest-path constraint, not a general ‘severe’ condition.
- [writing] The paper claims the benchmark ‘tests’ bi-directional anticipation, yet results show models fail to use backward retrieval (Section 4). The benchmark exposes a lack of this capability rather than testing a functional one. Rephrase to state the benchmark ‘reveals the inability of agents to utilize bi-directional planning’.
- [writing] The claim that failures are ‘most common’ when errors lack explicit signals conflates severity with frequency. While implicit failures cause lower accuracy, the paper does not quantify their prevalence in the failure set. Support the claim with frequency data or rephrase to focus on the severity of silent failures.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents a benchmark for evaluating LLM agents in large-scale tool ecosystems, focusing on planning under unreliable tool access. From a safety and ethics perspective, the paper is generally sound but requires specific clarifications to fully address data privacy, validation rigor, and potential dual-use risks.

First, the Ethics Statements section (located after the abstract) asserts that the retail-focused benchmark is unlikely to contain offensive material and that data was “sampled and validated.” However, it lacks the necessary granularity to verify this claim. The authors should explicitly detail the validation protocol: Was there a human review step? What were the specific criteria for “offensive content” or “PII” in a retail context? How many samples were reviewed? Without these details, the claim of “minimal societal risk” remains unsubstantiated. This is a standard requirement for datasets involving generated or curated text.

Second, regarding Data Privacy and Model Usage, the paper notes the use of proprietary models (GPT-5.4, Gemini) via APIs and local open-source models. When evaluating agents

that process queries and tool states, there is an implicit risk of data leakage to third-party API providers. The authors should add a statement confirming that the benchmark queries do not contain real-world Personally Identifiable Information (PII) and that the synthetic nature of the data mitigates privacy risks when sent to external APIs. If any real-world data was used for construction, a formal IRB or data usage agreement statement is required.

Third, while the current application is in the retail domain, the methodology involves simulating “blocking,” “implicit failures,” and “misleading tools.” This framework has potential dual-use implications. If adapted to critical infrastructure (e.g., financial trading, medical diagnosis, or industrial control systems), the ability to test how agents handle manipulated or failing APIs could be used to identify vulnerabilities in real-world systems. The authors should include a brief discussion in the Limitations or Ethics section acknowledging this potential for misuse and emphasizing that the benchmark is intended for defensive robustness testing only.

Finally, the paper mentions that “All annotations were performed by the authors.” If these annotations involved human judgment on the quality or safety of the generated tasks, the authors should briefly mention the qualifications of the annotators and any inter-annotator agreement metrics to ensure the safety validation is rigorous.

Addressing these points will strengthen the paper’s ethical standing and ensure transparency regarding data handling and potential risks.

Action items.

- [writing] The ‘Ethics Statements’ section (Section ‘Ethics Statements’) claims data was ‘sampled and validated’ to avoid offensive content but lacks specific details on the validation protocol, sample size, or human-in-the-loop review process. Explicitly describe the annotation guidelines and validation steps taken to ensure the retail dataset contains no PII or harmful content.
- [writing] The ‘Model Usage’ subsection notes the use of proprietary APIs (GPT-5.4, Gemini) and local open-source models. Clarify the data privacy implications of sending user queries and intermediate tool states to third-party API providers, and confirm that no sensitive user data is being processed or stored by these external services.
- [writing] The benchmark simulates ‘blocking’ and ‘noisy tools’ (Section 3.2) to test robustness. While currently in a retail domain, explicitly address the potential for dual-use if this framework were adapted to simulate attacks on real-world financial or medical systems (e.g., testing how agents handle manipulated API responses in critical infrastructure).

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling benchmark for long-horizon planning under retrieval constraints, supported by a substantial dataset (327 tasks, 1,665 tools) and a rigorous experimental setup involving 10 diverse LLMs. The use of bootstrap resampling (10,000 iterations) to generate confidence intervals for the primary metrics (Table 2) is a strong methodological choice that enhances the reliability of the reported performance bounds. The error analysis, particularly the categorization of “Irrecoverable Drift,” offers valuable qualitative evidence regarding agent failure modes.

However, the statistical evidence supporting the correlation claims in Section 4.2 requires strengthening. The authors report strong Pearson correlations (e.g., $r=0.902$ between Mean EDT and Accuracy) based on a sample size of $N=10$ (one data point per model). In statistical terms, a correlation coefficient derived from only 10 observations is highly sensitive to outliers and lacks the power to definitively establish a linear relationship without reporting p-values or confidence intervals for the correlation itself. While the trend is visually plausible, the claim of a “strong correlation” is statistically tenuous at this sample size. The authors should either provide the p-values and confidence intervals for these correlation coefficients or temper the language to reflect the exploratory nature of these findings given the limited N .

Additionally, the definition of “progress calls” in the error analysis (Section 6.1) relies on the existence of a ground-truth path. The paper mentions that tasks are constructed to have solvable paths, but it does not explicitly address how the metric handles cases where an agent discovers a *valid* alternative path that differs from the specific ground-truth path used for annotation. If an agent successfully solves a task via a novel path not in the original ground-truth set, but the metric penalizes steps that deviate from the specific ground-truth sequence, the “progress” metric could be biased. Clarifying the handling of alternative valid solution paths in the error annotation protocol is necessary to ensure the validity of the “Irrecoverable Drift” conclusions.

Finally, while Table 2 provides 95% confidence interval radii, the raw standard deviations or variances for the accuracy metrics in Table 3 are not explicitly listed. Including these would allow for a more direct assessment of the statistical significance of the performance gaps between top-performing models (e.g., the ~25% gap between Gemini-3.1-Pro and GPT-5.4) and would complement the bootstrap analysis.

Action items.

- [science] Clarify the statistical basis for the ‘strong correlation’ claims (e.g., $r=0.902$ between Mean EDT and Accuracy). The analysis compares only 10 data points (one per model). A Pearson correlation on $N=10$ is highly susceptible to outliers and lacks statistical power. Please report p-values, confidence intervals for the correlation coefficients, or justify the claim with a more robust statistical test.
- [science] Address the potential circularity in the ‘progress call’ definition used for error analysis. A call is ‘progress’ if it produces a value needed by a ground-truth path. However, if the ground-truth path is not unique or if the agent finds a valid alternative path

not in the original ground-truth set, this metric may penalize valid recovery. Explicitly state how alternative valid paths are handled in the error annotation.

- [science] Provide the standard deviation or variance for the accuracy metrics in Table 3, not just the 95% confidence interval radius. While the radius is provided in Table 2, the raw variance is necessary to assess the stability of the performance differences between models (e.g., the gap between Gemini-3.1-Pro and GPT-5.4) and to verify if the differences are statistically significant beyond the bootstrap intervals.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis in this paper is generally robust in its use of bootstrapping for confidence intervals (Table 1, 10,000 iterations), which appropriately addresses the uncertainty in the primary accuracy metrics given the sample size of $N=327$ tasks. However, several secondary statistical claims require further rigor to support the paper’s conclusions.

First, in Section 4.2, the authors report Pearson correlation coefficients between model performance and various behavioral metrics (e.g., $r = 0.902$ between Mean EDT and Accuracy). These correlations are computed over only $N = 10$ data points (the ten evaluated models). While the coefficients are high, the paper does not report confidence intervals, p-values, or standard errors for these correlations. With such a small sample size, the point estimates are highly volatile, and the claim that these relationships are “strong” lacks statistical precision. The authors should either provide the confidence intervals for these correlations or temper the language to reflect the exploratory nature of these findings.

Second, the error analysis in Section 6.2 presents detailed percentages for failure categories (e.g., “Irrecoverable Drift accounts for 72.4% of reported failure categories”). The statistical unit of analysis is ambiguous. It is unclear if these percentages are calculated over the total number of *failed trajectories* or the total number of *tool calls* within those trajectories. If the latter, the independence assumption of the data points is violated, potentially inflating the apparent prevalence of certain error types. The denominator for these proportions must be explicitly stated to ensure correct interpretation.

Finally, while the paper uses confidence intervals to show the spread of accuracy scores, it relies on visual inspection to claim that performance drops “sharply” under blocking conditions (Section 5, Figure 5). To strengthen the claim that blocking causes a statistically significant degradation in performance, the authors should perform a formal hypothesis test (e.g., a paired t-test or Wilcoxon signed-rank test) comparing the no-block and block accuracies for each model. The current presentation of confidence intervals for the means does not directly test the difference between the two conditions.

Action items.

- [science] Report confidence intervals or standard errors for the Pearson correlation coefficients ($r=0.902, 0.800, 0.781, -0.443$) in Section 4.2. With $N=10$ models, these point estimates lack precision metrics, making it difficult to assess the statistical significance of the claimed ‘strong’ relationships.
- [writing] Clarify the statistical unit of analysis for the error category percentages (e.g., 72.4% Irrecoverable Drift). Are these proportions calculated over the total number of failed trajectories or the total number of tool calls? The denominator must be explicit to interpret the magnitude of the failure modes correctly.
- [science] The claim that accuracy drops ‘sharply’ under blocking (Section 5) relies on visual inspection of Figure 5. Provide a formal statistical test (e.g., paired t-test or Wilcoxon signed-rank) comparing the no-block vs. block accuracies for each model to substantiate the significance of the performance degradation beyond the reported confidence intervals.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript is generally well-written, with a clear structure and logical flow. The abstract effectively summarizes the contributions, and the introduction sets up the problem well. However, there are several areas where the writing can be improved for clarity and conciseness.

In Section 4.1, the sentence structure regarding proprietary models is slightly awkward due to the unnecessary comma and the use of citation keys that may not be standard. This should be corrected to ensure smooth reading. Additionally, in Section 3.1, the repetition of “LLM” in the description of the datatype inventory process can be streamlined for better flow.

Section 5.1 contains a few phrases that could be rephrased for better readability, such as “Viable-path reduction sharply weakens performance.” The transition between defining categories in Section 6.2 and the subsequent analysis could also be smoother to maintain the reader’s engagement.

Overall, the paper is strong, but these minor revisions will enhance its clarity and professionalism.

Action items.

- [writing] In Section 4.1 (Settings), the text states ‘Proprietary models include GPT~\cite{GPT-5.4}, and Gemini~\cite{Gemini-3.1-Pro}’. The comma before ‘and’ is unnecessary and disrupts the flow. Additionally, the citation keys ‘GPT-5.4’ and ‘Gemini-3.1-Pro’ appear to be placeholders or non-standard; ensure these resolve to

valid bibliography entries.

- [writing] In Section 3.1 (Environment Setup), the sentence ‘The initial datatype inventory is proposed by a generation LLM M_{gen} and then automatically filtered by another LLM M_{fil} ’ is repetitive. Consider varying the sentence structure or combining the clauses for better flow.
- [writing] In Section 5.1 (Analysis), the phrase ‘Viable-path reduction sharply weakens performance’ is slightly awkward. ‘Reduction in viable paths’ or ‘Fewer viable paths’ would read more naturally. Also, ensure consistent capitalization in figure references (e.g., ‘Figure~\ref{fig:noise_combined}’ vs ‘the right panel of Figure...’).
- [writing] In the Abstract, the phrase ‘massive-tool planning remains challenging’ is clear, but the subsequent sentence ‘Failures are most common when errors lack explicit signals or when recovery demands longer alternative paths’ is a bit dense. Consider splitting this into two sentences or using a colon to improve readability.
- [writing] In Section 6.2 (Blocked-Alternative Misuse), the list items ‘Unused’, ‘Search Reused’, and ‘Value Reused’ are defined, but the subsequent paragraph ‘Semantically misleading tools are usually not the main issue’ lacks a clear transition from the previous definition. Ensure the logical flow between the definition of categories and the analysis of their frequency is smooth.